

aiSTROM instructor guide

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Use these materials to teach the aiSTROM roadmap (Herremans, 2021) or to facilitate an AI strategy workshop in an organisation. Participants work with the fillable worksheets; they do not need to read the paper first, although Sections III-A to III-H make good preparatory reading for students.

[Public guide](#) · [Blank worksheets](#) · [Completed expense-claims example](#) · [Paper](#)

Learning outcomes

Participants should be able to:

- Generate AI opportunities from efficiency gaps, new technologies, competitors and new business models, and shortlist the top n by impact and effort.
- Analyse each shortlisted project for data, AI team, positioning in the company, technologies, KPIs and risk level.
- Explain why the same question (e.g. “do we have the data?”) can lead to very different answers for different projects.
- Weigh risks against benefits using a risk-benefit plot and SWOT, and build a portfolio that fits an organisation’s risk strategy.
- Plan the education and cultural shift that adoption requires.

How the worksheets work

Sheet	Paper	Activity
Cover		Team, organisation and starting situation.
1. Goals	III-A	List up to 8 ideas, score impact and effort, shortlist A, B, C.
2-7	III-B to III-G	One sheet per roadmap step, one column per project, a conclusion row at the bottom.
8. Cultural shift	III-H	Education and adoption for each project.
9. Decide	III-G.3	Risk-benefit plot, SWOT per project, decision per project and portfolio.

Because every sheet has a column per project, groups compare projects row by row. That comparison is the heart of the method: a project that looks attractive on its own often looks different next to the alternatives.

Preparation

Give each group a blank set of worksheets (printed at A3, or as fillable PDFs) and a scenario: a real organisation with shareable information, or one of the exercises in this guide. Mix business, domain and technical backgrounds in each group, as the paper recommends for Step 1. Ask participants not to enter confidential information into shared files or unapproved AI services.

Show the completed example **after** groups have made their own shortlist, so they don’t copy it.

A 90-minute session

Time	Activity	Output
0-10 min	Introduce the roadmap (cover sheet) and the scenario.	Shared understanding of the situation.
10-25 min	Sheet 1: brainstorm ideas from the four sources, score impact and effort, choose the top three.	Shortlist A, B, C with a reason.
25-55 min	Sheets 2-7: split the group; each pair takes two sheets and fills them for all three projects.	Filled sheets with conclusions.
55-65 min	Pairs report their conclusion rows. Which sheet changed the ranking?	Shared view of differences.
65-80 min	Sheet 9: place projects on the risk-benefit plot, SWOT, decide per project.	Portfolio decision and next steps.
80-90 min	Debrief with the completed example; mention sheet 8 as homework.	Reflection.

For a half-day workshop, work through sheets 2-8 together, one per round, and let the groups present their portfolio to a “board” played by another group. For a university course, assign one roadmap step per week alongside the matching paper section.

Facilitation prompts

- **Goals:** Is this a clear problem statement, or a technology looking for a problem? Does it automate processing, find trends or make predictions?
- **Data:** Do you already have it? Would more data really help? Which privacy laws apply in every country you operate in?
- **AI team:** Who brings the domain knowledge? Up-skill, hire or acquihire?
- **AI in the company:** Does a service already exist (don't reinvent the wheel)? What prototype would test the idea cheaply?
- **Technologies:** Must the result be explainable? Does the system replace or augment people? Where do the labels come from?
- **KPIs:** Is the KPI about value, or only about model accuracy? When would you pivot?
- **Risk level:** What if the model simply doesn't work? Who could be treated unfairly? How could someone fool it?
- **Decide:** Does the portfolio match the organisation's risk strategy? What unlocks the next funding round?

Discussing the expense-claims example

The [completed example](#) follows a fictional 1,200-person company that shortlists three projects: receipt auto-fill (A), fraud and duplicate flags (B) and a policy pre-check (C). Useful questions:

1. **Same question, different answers.** Compare the Data sheet for A and B. Why does B's real first step become data collection?
2. **Explainability.** Why does B need an interpretable model while A does not? Link to the paper's credit-scoring example.
3. **Not everything needs AI.** C ends up as a rules engine, and the spend dashboard (idea 4) goes to the BI team. Is that a failure of the AI strategy, or a success?
4. **Adversarial inputs.** AI-generated fake receipts appeared widely in 2025. Which sheets change because of this, and which project benefits?
5. **Portfolio.** Would a more risk-seeking company decide differently? What would it need to believe?

Assessing the work

Assess reasoning rather than filled boxes. Look for: a credible shortlist with reasons; real differences between projects in the columns; conclusions that follow from the rows; a decision per project that matches the risk-benefit plot; and next steps with an owner and a date.

Workshop exercises

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All scenarios and figures are fictional. There is no single correct answer; good answers are well reasoned and consistent across the sheets.

Exercise A: a manufacturer

A manufacturer of packaging materials (800 employees, three plants) wants an AI strategy. Facts the group can use:

- Two years of sensor readings from production lines, with repair notes and downtime records. Machine identifiers differ between systems, several sensors were replaced, and failure labels are incomplete.
- Quality inspectors visually check samples of every batch; about 2% of batches are rejected late.
- Sales forecasts are made in spreadsheets by three planners.
- A competitor advertises “AI-optimised” delivery times.
- There is one data analyst; no AI specialists.

Task: run the full roadmap. Generate ideas on sheet 1 (expect predictive maintenance, visual quality inspection, demand forecasting and others), shortlist three, analyse them on sheets 2-7 and decide on sheet 9.

Change halfway through: the maintenance lead says the team can handle only a few extra inspections per week. What does that mean for the KPIs and model metrics of predictive maintenance (precision versus recall)?

Debrief: visual inspection is an established task with off-the-shelf services; predictive maintenance depends on labels the company barely has. Did groups notice? Did anyone propose a hybrid with simple rules?

Exercise B: expense claims

Use the cover sheet of the completed example as the scenario, but give groups the **blank** worksheets. Let them generate their own ideas before showing the example.

Change halfway through: news breaks that employees elsewhere are submitting AI-generated fake receipts, realistic down to the coffee stains. Which project gains value? Which project gains risk? Does the portfolio change?

Debrief: compare the groups' shortlists and portfolios with the completed example. Differences are fine if the reasoning holds.

Exercise C: a public service

A city council (2,000 staff) receives 40,000 citizen emails and forms a year: permit questions, complaints, requests. Ideas on the table include routing messages to the right department, a chatbot for permit questions, and predicting which streets need maintenance from complaint data.

Task: shortlist and analyse. Pay special attention to explainability (citizens are entitled to reasons), bias (some neighbourhoods complain more than others) and accountability for chatbot answers. Discuss *Moffatt v. Air Canada* (2024), in which an airline was held responsible for a refund policy its chatbot invented.

Exercise D: challenge another group

Swap the Decide sheet with another group. Pick the project they decided to pursue first and find the weakest row on sheets 2-7 behind it. Ask for the evidence. Then propose one prototype or data step that would reduce that uncertainty, with an owner and a date.